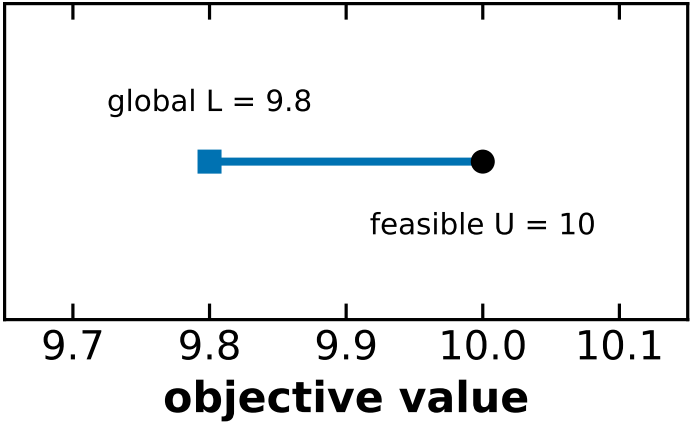

Interval width: 0.2

A local first-order residual answers a different question.

Global bounds must cover the full feasible domain.